

End-to-End Loan Data Analytics

From BI Dashboards to Predictive Machine Learning

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Last Update: March 25, 2026

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Current Projects & Roles

- **Data Analyst (Student Council):** Leveraging data to drive student-centric initiatives at Chulalongkorn University.
- **OR Scheduling Optimization:** Developing data-driven models to optimize doctor schedules in Operating Rooms (Side Project).
- **CTO, Simulated Enterprise (Core Course):** Elected by peers to co-manage "IBM Company," a simulated trading business, alongside the executive team.
- **Freelance Writer:** Independent film critic and content creator.

Executive Summary

🎯 Project Objective & Scope

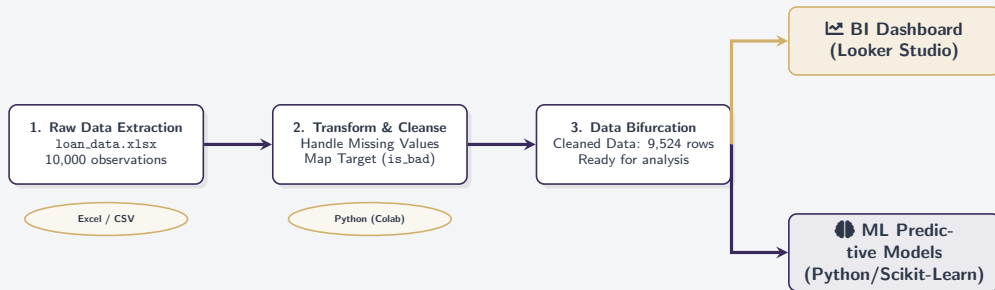
To engineer a data pipeline that transforms 10,000 raw loan records into actionable business intelligence, and deploy machine learning models for proactive credit risk mitigation.

- **Data Architecture:** Cleaned data and engineered features, yielding a high-quality dataset of 9,524 records.
- **Descriptive Analytics (Looker Studio):** Uncovered a base portfolio default rate of **4.74%**. Identified the *Small Business* segment as a severe vulnerability with a **12.24%** default rate.
- **Predictive Analytics (Machine Learning):** Utilized Classification models (SVM) to predict defaults and Clustering (DBSCAN) to detect anomaly profiles like "High-Income but Maxed-Out Debt."
- **Strategic Action:** Propose dynamic risk-based pricing and an automated API gatekeeper to filter high-risk applicants.

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1. End-to-End Data Pipeline Architecture



*This pipeline ensures that **Dashboards (Looker)** and **Machine Learning** models share the exact same clean data.*

[!\[\]\(bd1a142de767a21e5362c595f844a4ff_img.jpg\) Click here to view the Data Dictionary \(Appendix\)](#)

2. Deep Dive: Data Transformation Process

To build a reliable Dashboard and ML model, we prepared the data through these key steps in Python:

1. Data Cleansing

Handling Missing Values: Removed rows where employment length (`emp_length`) was 'n/a'.

- **Why?** Borrowers without employment history distort risk profiles. Removing them gave us a clean dataset of 9,524 records.

2. Target Variable Definition

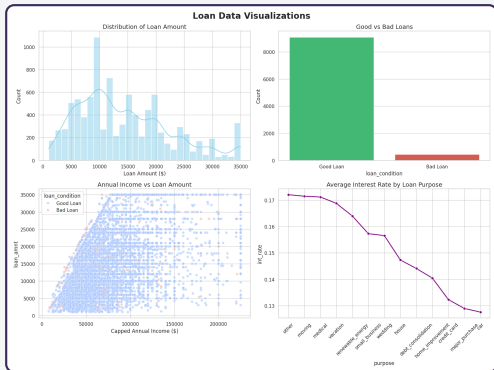
Grouped 7 complex `loan_status` categories into a simple **Binary Target** (`is_bad`) for ML to learn from:

- **Class 1 (Risk):** Default, Charged Off, Late.
- **Class 0 (Safe):** Current, Fully Paid.

3. Feature Engineering

Transformed raw text (like `small_business`) to clean labels. Grouped numerical Debt-to-Income (`dti`) into Low, Moderate, and High buckets to make insights easier to read on the dashboard.

3. Exploratory Data Analysis (EDA)



Visualizing Feature Distributions (Python)

Pre-Modeling Insights

Before training the AI, we explored the data patterns:

- **Data Scaling:** Variables like Income and Loan Amount have vastly different ranges. We applied scaling techniques so the ML models evaluate all features equally.
- **Correlation:** Interest Rate (`int_rate`) and Debt Burden (`dti`) show strong positive trends toward loan default.

Analytical Takeaway

High Interest Rate is not the *cause* of default; it is a *symptom* of a borrower already having a high Debt-to-Income burden.

Loan Portfolio Performance

int_rate_ca...

term

home_ownership

KPIs Summary

Total Applications

9,524

Total Funded Volume

฿136,380,700.00

Outstanding Principal

฿97,655,993.36

Total Interest Received

฿36,272,166.49

Portfolio Default Rate

4.74%

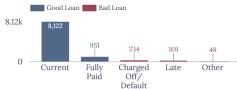
Portfolio Analysis

Portfolio Health

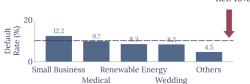


● Good Loan
● Bad Loan

Detailed Loan Status Breakdown



Risk Segment by Exposure



Risk Deep Dive

Interest Rate by Debt Burden



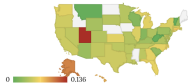
Average Interest Rate by Risk Segment



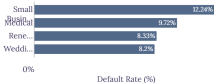
Employment Length vs. Default Risk

Empl...	Home Ow...	Defau...
1. 08: 8 Years	OWN	13.64%
2. 01: < 1 Year	OWN	10.91%
3. 02: 1 Year	OWN	10%
4. 05: 5 Years	RENT	7.67%
5. Other	OWN	7.44%
6. 08: 8 Years	RENT	7.18%
7. Other	RENT	6.74%
8. 06: 6 Years	MORTGAGE	6.46%
9. 06: 6 Years	RENT	6.3%

State Risk Distribution



Loan Purpose



1 - 35 / 35

4. Deep Dive: Looker Studio Executive Insights

Open Live Looker Studio Dashboard

- **The SME Vulnerability (12.24% Default Rate):** While the portfolio average default rate rests at 4.74%, the *Small Business* sector is underperforming.
Context: SMEs suffer from high cash flow volatility and lack robust collateral [1], making them highly susceptible to economic shocks.
- **Debt-to-Income (DTI) as a Constraint:** Borrowers exceeding a 20% DTI threshold dominate the delinquency segments.
Context: High DTI signifies a lack of personal liquidity. Any unexpected life event translates immediately into loan defaults [2].
- **Risk-Based Pricing Misalignment:** The interest rate gap between Safe Loans (13.47%) and Defaulted Loans (14.92%) is merely **1.45%**. This small gap is insufficient to cover potential losses [3].

5. Supervised Learning: SVM Classification Engine

Access Full Interactive Google Colab Environment

Algorithmic Logic: Support Vector Machine (SVM)

We deployed a Classification model (SVM) to predict whether a new applicant will be a **Good** or **Bad** loan. We utilized an automated tuning process (GridSearchCV) to find the best settings, ensuring the model is highly accurate without merely memorizing the data (Overfitting).

```
# Hyperparameter Tuning for Support Vector Machine (SVM)
from sklearn.svm import SVC
from sklearn.model_selection import GridSearchCV

# GridSearch tests multiple settings to find the optimal model configuration
param_grid = {'C': [0.1, 1, 10], 'gamma': ['scale', 'auto'], 'kernel': ['rbf']}
grid = GridSearchCV(SVC(), param_grid, refit=True, cv=5)
grid.fit(X_train_scaled, y_train)

print(f"Optimal Model Parameters: {grid.best_params_}")
```

6. Unsupervised Learning: Anomaly Detection

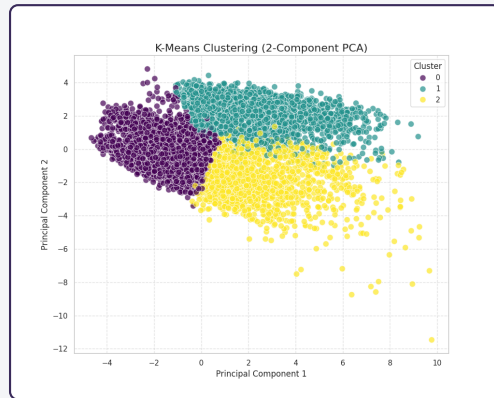
Clustering Logic

We used **DBSCAN** clustering to automatically identify natural groupings and detect "Noise" (Outliers) based on similarities in borrowers' financial behaviors.

Hidden Risk Discovered

The model flagged an anomalous group: *"High Income, but Maxed-out Revolving Balance"*.

Traditional rules approve high-income applicants quickly. However, ML proves that if their credit cards are maxed out, their risk of default actually surges.



Isolating Anomalous Noise Points via DBSCAN

7. Strategic Implementation & Conclusion

By synthesizing retrospective BI analytics with predictive Machine Learning, we propose the following actionable steps:

- 1 Dynamic Risk-Based Pricing Structure:** Calibrate the pricing model. Increase the risk premium by at least 3-4% for SME applicants and individuals exceeding a 20% DTI threshold to accurately provision for expected losses.
- 2 Automated ML Gatekeeper (API Deployment):** Export the tuned SVM model and integrate it directly into the Loan Origination System. This calculates an applicant's Probability of Default (PD) in real-time before funding is approved.
- 3 Policy Revision on Employment Tenure:** Update credit policies to cap loan amounts for borrowers with less than 2 years of employment, as the data strictly penalizes early-career income instability.

8. References

-  Board of Governors of the Federal Reserve System.
Report to the Congress on the Availability of Credit to Small Businesses, 2022.
<https://www.federalreserve.gov/publications/report-to-the-congress-on-the-availability-of-credit-to-small-businesses.htm>
-  Consumer Financial Protection Bureau (CFPB).
Data Point: Debt-to-Income and Loan Performance, 2023.
<https://www.consumerfinance.gov/data-research/research-reports/data-point-debt-to-income-and-loan-performance/>
-  Basel Committee on Banking Supervision.
Principles for the Management of Credit Risk, 2020.
<https://www.bis.org/bcbs/publ/d502.htm>

Appendix A: Data Field Definitions (Part 1/2)

Feature Name	Description
id	Borrower's unique identification number
loan_amnt	Amount of loan applied for
funded_amnt	Outstanding principal amount
term	Term of loan
int_rate	Interest rate
installment	Monthly payment
emp_length	Borrower's employment length
home_ownership	Borrower's home ownership
annual_inc	Borrower's annual income
loan_status	Status of the loan (Target Variable)
purpose	Purpose of the loan
addr_state	Borrower's residence state

Appendix A: Data Field Definitions (Part 2/2)

Feature Name	Description
dti	Borrower's debt-to-income ratio
delinq_2yrs	Frequency of the borrower's delinquencies in the last 2 years
earliest_cr_line	Date the borrower started his/her first credit line
mths_since_last_delinq	Number of months since the borrower's last delinquency
open_acc	Number of the borrower's open credit accounts
revol_bal	Borrower's revolving balance
total_acc	Number of total credit accounts (open & closed)
out_prncp	Outstanding principal of the loan
total_pymnt	Total payment on the loan
total_rec_prncp	Total received principal on the loan
total_rec_int	Total received interest on the loan

Thank You

Data Analytics

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